**Questio Paper Code: 25901225**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCECT201 - AI FOR SMARTELECTRONICS

(2024 ADMISSION ONWARDS)

**Duration: 1 hour 30 minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)***

**ANSWER KEY**

### Part A

MCQ

Answer all questions

Each question carries **1** marks

| | | |
|---|---|---|
| 1. | Identify the primary function of a virtual assistant.<br><br>c). User interaction and taskassistance | [U] / [CO1] |
| 2. | Define unsupervised learning.<br><br>c). Learning without labeled data | [K] / [CO2] |
| 3. | Identify from option,the storage requirement is measured in<br><br>c). Bytes | [K] / [CO3] |
| 4. | Select the primary goal of mathematical optimization in AI.<br><br>c). Finding the best solution under constraints | [U] / [CO1] |
| 5. | Name the basic unit of a neural network.<br><br>b). Neuron | [K] / [CO2] |
| 6. | Why are face apps and customized object detection considered basic real-worldapplications of computer vision?<br><br>d). They apply visual analysis to solve practical problems | [An] / [CO3] |
| 7. | In navigation applications, the "Shortest Path Problem" refers to:<br><br>b). Minimizing a cost function (timeor distance) between two nodes | [U] / [CO1] |
| 8. | Select the learning type that combines labeled and unlabeled data.<br><br>d). Semi-supervised | [U] / [CO2] |
| 9. | Select the correct CAD applications requirement<br><br>b). High GPU | [U] / [CO3] |
| 10. | Name the process of improving model accuracy using past errors.<br><br>c). Training | [K] / [CO2] |

**[1x10 = 10]**

### Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Identify one limitation that both Narrow AI and Generative AI share, whichGeneral AI would theoretically overcome. [U] / [CO1]

- Both Narrow AI and Generative AI lack true understanding and general intelligence.
- They are task-specific and cannot reason or adapt across domains independently.
- Depend heavily on training data and may fail in unfamiliar situations.

General AI (AGI) would overcome this by:

- Exhibiting human-like reasoning and adaptability
- Performing multiple tasks across domains without retraining

12. Define classification and regression. [K] / [CO2]

- Classification:
  A supervised learning method that assigns data into discrete categories
  (e.g., spam vs non-spam emails)
- Regression:
  A supervised learning method used to predict continuous numerical values
  (e.g., predicting house prices)

Key Difference:
Classification → categorical output
Regression → numerical output

13. Why is camera resolution an important factor while selecting hardware forcomputer vision. [An] / [CO3]

- Higher resolution provides more detail and clarity
- Improves object detection accuracy
- Helps in recognizing small or distant objects
- Reduces errors in image processing and analysis
- Enhances model performance and reliability

14. Classify the main approaches to simulating the human brain, such as bottom-upand top-down methods. [U] / [CO1]

- Bottom-Up Approach:
  - Builds intelligence from basic units (neurons)
  - Example: Artificial Neural Networks
  - Focus on learning from data
- Top-Down Approach:
  - Uses rules, logic, and symbolic reasoning
  - Example: Expert systems
  - Focus on human knowledge representation

Classification:

- Bottom-up → Data-driven
- Top-down → Rule-based

15. List the main components of a reinforcement learning system. [K] / [CO2]

- Agent: Learner/decision maker
- Environment: Where the agent operates
- Actions: Choices made by the agent
- State: Current situation of the agent
- Reward: Feedback signal for actions

16. Show how AI upscaling can improve user experience in streaming services. [U] / [CO1]

- AI upscaling uses machine learning models to enhance low-resolution videos
- Converts SD/HD content into near 4K quality
- Improves sharpness, textures, and details
- Reduces blurriness and noise
- Provides better viewing experience on platforms like Netflix

Impact:

- Better quality without higher bandwidth
- Enhanced user satisfaction

**[5x4 = 20]**

## Part C

Essay Questions

Answer any **2** questions
Each question carries **10** marks

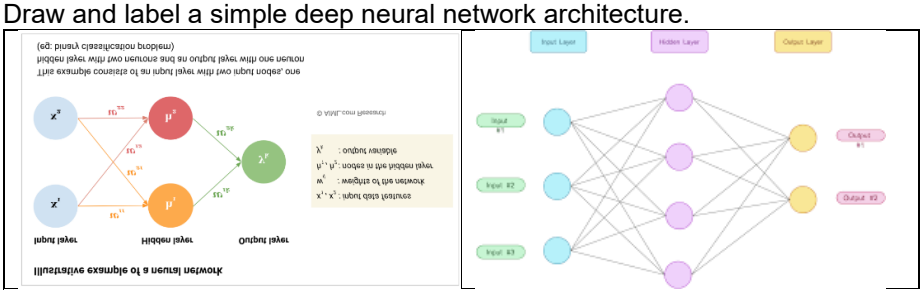
17. Compare Artificial Intelligence with human intelligence in terms of learning and reasoning. [U] / [CO1]

| Aspect | Artificial Intelligence | Human Intelligence |
|---|---|---|
| Learning | Learns from data and algorithms | Learns from experience, emotions, and reasoning |
| Reasoning | Logical and rule-based | Flexible, intuitive, and creative |
| Adaptability | Limited to training data | Highly adaptable to new situations |
| Speed | Very fast processing | Slower but more thoughtful |
| Creativity | Limited (pattern-based) | Highly creative and imaginative |

18. Discuss the role of neural network architecture in smart electronic systems. [U] / [CO2]

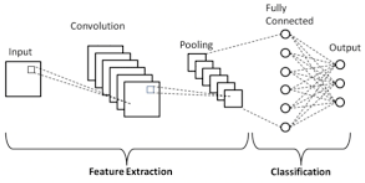
- Determines how data flows through the system (input → hidden → output layers)
- Influences accuracy and efficiency of the system
- Enables pattern recognition (e.g., speech, images)
- Supports automation in smart devices (IoT, robotics, smart homes)
- Architecture choice (CNN, RNN, etc.) affects performance and application suitability
  A well-designed neural network architecture is essential for reliable and intelligent system behavior.

19. Draw and label a simple deep neural network architecture. [K] / [CO2]



- Input Layer: Receives raw data and determines the input dimension, with nodes equal to input features.
- Hidden Layers: One or more layers (e.g., Dense/Fully Connected) containing neurons that compute a weighted sum of inputs and apply non-linear activations.
- Output Layer: Produces the final prediction, such as a single value for regression or multiple probabilities for classification (e.g., Softmax).
- Weights and Biases: Learnable parameters connected to each node that are updated through training to minimize error.

20. Illustrate the working of Convolutional Neural Networks (CNNs) in computervision applications with a neat diagram. [An] / [CO3]




Step-by-step Working:

1. Input Image: Raw image data
2. Convolution Layer: Applies filters to extract features (edges, textures)
3. ReLU Activation: Introduces non-linearity
4. Pooling Layer: Reduces size (downsampling)
5. Fully Connected Layer: Combines features

6. Output Layer: Classification result (e.g., cat/dog)

Applications:

- Face recognition
- Object detection
- Medical image analysis

CNNs automatically extract important features from images, making them highly effective for computer vision tasks.

**[10x2 = 20]**